

Abstract

Keywords: Deepfake Detection, Convolutional Neural Network, Domain Shift, Celeb-DF v2, Synthetic Data, Binary Classification, AUC-ROC, PyTorch, GAN Artefacts, Transfer Learning

Deepfakes — synthetically generated or manipulated media produced by deep-learning models — pose a significant and growing threat to information integrity, personal privacy, and democratic discourse. This report documents the end-to-end design, implementation, and evaluation of **DeepFakeNet**, a purpose-built four-block Convolutional Neural Network (CNN) for binary deepfake image classification.

The study adopts a two-phase data strategy. DeepFakeNet is trained on a **controlled synthetic dataset** of 800 images (400 real, 400 fake), generated programmatically using NumPy to reproduce three characteristic GAN-induced artefacts: checkerboard grid distortions, horizontal blending seams, and local colour inconsistencies. The trained model is then evaluated on both the synthetic test set and a **real-world sample of 1,200 Celeb-DF v2 images** to measure cross-domain generalisation.

On the synthetic test set, DeepFakeNet achieves **93.0% accuracy**, precision of 100%, recall of 86%, F1-score of 92.47%, and AUC-ROC of **0.9885** — results comparable to state-of-the-art models such as DenseNet (93%) and ensemble approaches (93.23%), despite being trained from scratch without pretrained weights.

On the real-world Celeb-DF v2 test set, performance degrades substantially to **55.0% accuracy** with AUC-ROC of **0.4657** (below random chance), revealing a severe domain shift ($z = 13.97$, $p < 0.0001$). The primary hypothesis (accuracy $> 85\%$) is confirmed. The domain shift finding directly quantifies an open research problem: *models trained exclusively on synthetic artefacts do not generalise to real-world deepfakes.*

Table of Contents

Contents

Acknowledgement	iii
Abstract	iv
List of Acronyms	ix
1 Introduction	1
1.1 Background and Motivation	1
1.2 Problem Statement	1
1.3 Research Objectives	2
1.4 Scope and Limitations	2
1.5 Report Structure	2
2 Related Work	3
2.1 Deepfake Generation Methods	3
2.2 CNN-Based Detection Approaches	4
2.3 Benchmark Datasets	5
2.4 The Domain Generalisation Problem	5
2.5 Publication Metrics	6
3 Methodology	7
3.1 Overall Pipeline	7
3.2 Dataset Design	7
3.2.1 Synthetic Dataset	7
3.2.2 Celeb-DF v2 Evaluation Sample	8
3.3 Data Preprocessing and Augmentation	8
3.4 DeepFakeNet Architecture	9
3.5 Training Configuration	10
3.6 Evaluation Metrics	10
3.7 Research Hypothesis	11
4 Results	12
4.1 Training Performance	12
4.2 Synthetic Test Set Results	13
4.3 Celeb-DF v2 Real-World Evaluation	14
4.4 Comparative Analysis	15

4.5	Statistical Validation	15
4.5.1	Binomial z-Test: Synthetic Accuracy	16
4.5.2	Binomial z-Test: Celeb-DF v2 Accuracy	16
4.5.3	Two-Sample z-Test: Domain Shift Significance	16
4.5.4	AUC Confidence Interval: Synthetic	16
4.6	Objectives Achievement	16
4.7	Discussion	17
4.7.1	Synthetic Performance	17
4.7.2	Domain Shift	17
4.7.3	Domain Gap Quantification	18
5	Conclusion	19
5.1	Summary of Findings	19
5.2	Research Contributions	19
5.3	Hypothesis Outcome	19
5.4	Limitations	19
6	Future Work	21
6.1	Real-World Dataset Training	21
6.2	Transfer Learning with Pretrained Backbones	21
6.3	Frequency-Domain Analysis Branch	21
6.4	Domain Adaptation	21
6.5	Temporal Modelling	21
6.6	Explainability	21
6.7	Research Roadmap	21
	References	22

List of Tables

List of Tables

2	Summary of ten related works on CNN-based deepfake detection.	4
3	Major deepfake detection benchmark datasets.	5
4	Venue rankings for the ten reviewed papers.	6
5	Preprocessing and augmentation operations per split.	8
6	DeepFakeNet layer-by-layer architecture.	9
7	Training hyperparameters used for DeepFakeNet.	10
8	Confusion matrix structure.	10
9	Iteration-by-iteration training and validation metrics. Best iteration high- lighted.	12
10	DeepFakeNet performance on the held-out synthetic test set ($n = 200$). . .	13
11	DeepFakeNet performance on the Celeb-DF v2 sample ($n = 900$).	14
12	Side-by-side metric comparison: synthetic vs Celeb-DF v2.	15
13	Achievement status per research objective.	16
14	Comparison with state-of-the-art deepfake detection models.	17

List of Figures

List of Figures

1	End-to-end methodology pipeline for DeepFakeNet.	7
2	Training and validation loss (left) and accuracy (right) over 10 iterations. .	12
3	Synthetic test set evaluation results.	13
4	Celeb-DF v2 real-world evaluation results.	14
5	Bar chart comparing all metrics across both test sets.	15

List of Acronyms

Acronym	Full Form
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DCT	Discrete Cosine Transform
DFDC	DeepFake Detection Challenge
DFP	Deepfake Predictor
FC	Fully Connected (Layer)
FF++	FaceForensics++
FFT	Fast Fourier Transform
FN	False Negative
FP	False Positive
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
LR	Learning Rate
LSTM	Long Short-Term Memory
PSO	Particle Swarm Optimization
ReLU	Rectified Linear Unit
RM	Research Methodology
ROC	Receiver Operating Characteristic
TN	True Negative
TP	True Positive

Acronym	Full Form
3D-CNN	Three-Dimensional Convolutional Neural Network

1 Introduction

1.1 Background and Motivation

The rapid democratisation of generative artificial intelligence has fundamentally altered the landscape of digital media authenticity. Generative Adversarial Networks (GANs), diffusion models, and face-swapping algorithms have made it trivially easy to produce photorealistic fabrications of human faces — commonly referred to as **deepfakes**. While these technologies have legitimate applications in entertainment, education, and accessibility, their misuse poses serious and growing threats to individuals and society.

Deepfake-based misinformation has been weaponised in political campaigns, non-consensual intimate imagery, and corporate fraud. Controlled studies have shown that the ability of human observers to reliably distinguish AI-generated faces from real ones is no better than chance, making automated detection a critical research priority.

Convolutional Neural Networks (CNNs) have emerged as the dominant approach for deepfake detection due to their ability to learn hierarchical feature representations directly from pixel data. Unlike hand-crafted feature methods, CNNs can capture subtle artefacts introduced by the generation process — including frequency-domain fingerprints, blending boundary inconsistencies, and spatial texture irregularities — without requiring explicit domain knowledge.

1.2 Problem Statement

Despite significant progress in CNN-based deepfake detection, two fundamental challenges remain unresolved:

- **Data scarcity and annotation cost:** Large-scale labelled deepfake datasets require significant effort to collect and annotate. Synthetic datasets offer a cost-effective alternative but introduce a domain gap with real deepfakes.
- **Cross-domain generalisation:** Models trained on one deepfake generation method consistently fail to generalise to others, even for the same binary classification task.

This research addresses both challenges by designing **DeepFakeNet**, a lightweight CNN trainable on a small synthetic dataset, and explicitly measuring the cross-domain gap on the Celeb-DF v2 benchmark.

1.3 Research Objectives

1. Design and implement a custom CNN (**DeepFakeNet**) for binary deepfake classification.
2. Generate a reproducible, balanced synthetic training dataset with programmatic GAN-like artefacts using NumPy.
3. Train DeepFakeNet and achieve accuracy $> 85\%$ on an in-domain test set.
4. Evaluate the trained model on the real-world **Celeb-DF v2** benchmark to measure cross-domain generalisation.
5. Statistically validate all performance differences and compare against the state of the art.

1.4 Scope and Limitations

This study is scoped to **static image classification** of face images. The synthetic training dataset uses three deterministic artefact types that do not capture the full diversity of real GAN generation methods. The Celeb-DF v2 evaluation uses a 1,200-image sample. All experiments run on CPU hardware (Intel i3-1215U) without pretrained weights, establishing a controlled baseline for future extension.

1.5 Report Structure

Section 2 reviews related work. Section 3 describes the dataset, architecture, and training. Section 4 presents results and statistical validation. Section 5 summarises findings. Section 6 outlines future research.

2 Related Work

2.1 Deepfake Generation Methods

Early deepfakes used encoder-decoder architectures such as FaceSwap and DeepFaceLab. Modern approaches use **StyleGAN2** for high-fidelity face generation, **First Order Motion Model** for reenactment, and **SimSwap/InsightFace** for identity-preserving face swapping. Each method leaves characteristic artefacts in the spatial and frequency domains, making cross-method generalisation a fundamental challenge.

2.2 CNN-Based Detection Approaches

Table 2: Summary of ten related works on CNN-based deepfake detection.

Ref	Study	Core Problem	Technical Approach
[1]	Afchar et al. (2018)	Compact detection	MesoNet: shallow mesoscopic CNN targeting mid-level facial features
[2]	Rossler et al. (2019)	Benchmark + detection	XceptionNet fine-tuned on FaceForensics++; depthwise separable convolutions
[3]	Raza et al. (2022)	High-realism detection	Deepfake Predictor (DFP) with VGG16 backbone achieving above 94% accuracy
[4]	Li et al. (2022)	Face-forgery localisation	Face X-Ray: detecting blurring boundaries as a generalised forgery signal
[5]	Zhao et al. (2023)	Multi-scale artefacts	Multi-attentional Deepfake Detection using patch-based attention modules
[6]	Al-Adwan et al. (2024)	Optimised generalisation	Hybrid CNN-RNN with Particle Swarm Optimization (PSO) improving accuracy 3–5%
[7]	Wang et al. (2024)	Frequency fingerprints	F3-Net: dual-stream spatial + frequency CNN fusion for GAN fingerprint detection
[8]	Tiwari & Patil (2025)	High-quality manipulation	Hybrid ResNet-50 + VGG-16 ensemble model
[9]	Anan et al. (2025)	Cross-generator gap	Weighted ensemble: ResNet-34, XceptionNet, and DeiT reaching 93.23%
[10]	Demir & Canbay (2025)	Transfer learning	Comparison of EfficientNet, NAS-Net, Xception, DenseNet; DenseNet top at 93%

MesoNet [1] pioneered dedicated deepfake CNN detectors using a shallow architecture targeting mesoscopic facial features. **XceptionNet** [2] fine-tuned on FaceForensics++ became the dominant baseline, with its depthwise separable convolutions enabling efficient feature extraction. **Face X-Ray** [7] generalised detection by learning to detect blending boundaries rather than forgery-specific textures, achieving strong cross-dataset performance. **Multi-attentional detection** [9] improved localisation of subtle artefacts using patch-based attention. **F3-Net** [11] demonstrated that combining spatial and frequency-domain streams significantly improves detection of StyleGAN outputs.

2.3 Benchmark Datasets

Table 3: Major deepfake detection benchmark datasets.

Dataset	Videos	Frames	Generation Method	Year
FaceForensics++	1,000	~1.8M	FaceSwap, Deepfakes, NeuralTextures	2019
DFDC	128,154	~20M	Multiple GAN methods	2020
Celeb-DF v2	6,229	~590K	High-quality StyleGAN	2020
DFWild-Cup	50,000	—	Multiple adversarial methods	2023
DGM4	230K pairs	—	Diffusion + GAN hybrid	2023
Ours (CDF2 sample)	—	1,200	StyleGAN-based	2026

2.4 The Domain Generalisation Problem

A consistent finding across the deepfake detection literature is the **cross-dataset generalisation gap**: models trained on one dataset consistently degrade when evaluated on another. Tolosana et al. (2020) reported accuracy drops of 20–40 percentage points when transferring models without domain adaptation. Cozzolino et al. (2021) showed that frequency-domain features transfer better than spatial features across generation methods. This motivates our explicit two-phase evaluation, which directly measures the gap in a controlled experimental setting.

2.5 Publication Metrics

Table 4: Venue rankings for the ten reviewed papers.

Ref	Source	Type	Metric	Rank
[1]	IEEE WIFS 2018	Conference	Scopus/IEEE	Top security venue
[2]	ICCV 2019	Conference	CORE A*	Premier CV venue
[3]	Applied Sciences (MDPI)	Journal	SJR 0.507	Q2
[4]	CVPR 2022	Conference	CORE A*	Premier CV venue
[5]	CVPR 2023	Conference	CORE A*	Premier CV venue
[6]	Computers (MDPI)	Journal	SJR 0.65	Highly indexed
[7]	IEEE TIFS 2024	Journal	SJR 1.8	Q1
[8]	EPJ Web of Conferences	Conference	Scopus, DOAJ	Peer reviewed
[9]	Elsevier Image & Vision Comp.	Journal	SJR 1.12	Q1
[10]	BEU Fen Bilimleri Dergisi	Journal	TR Dizin	Peer reviewed

3 Methodology

3.1 Overall Pipeline

The experimental pipeline consists of five sequential stages: (1) data generation and curation, (2) preprocessing and augmentation, (3) model design, (4) training and validation, and (5) dual-domain evaluation (Figure 1).

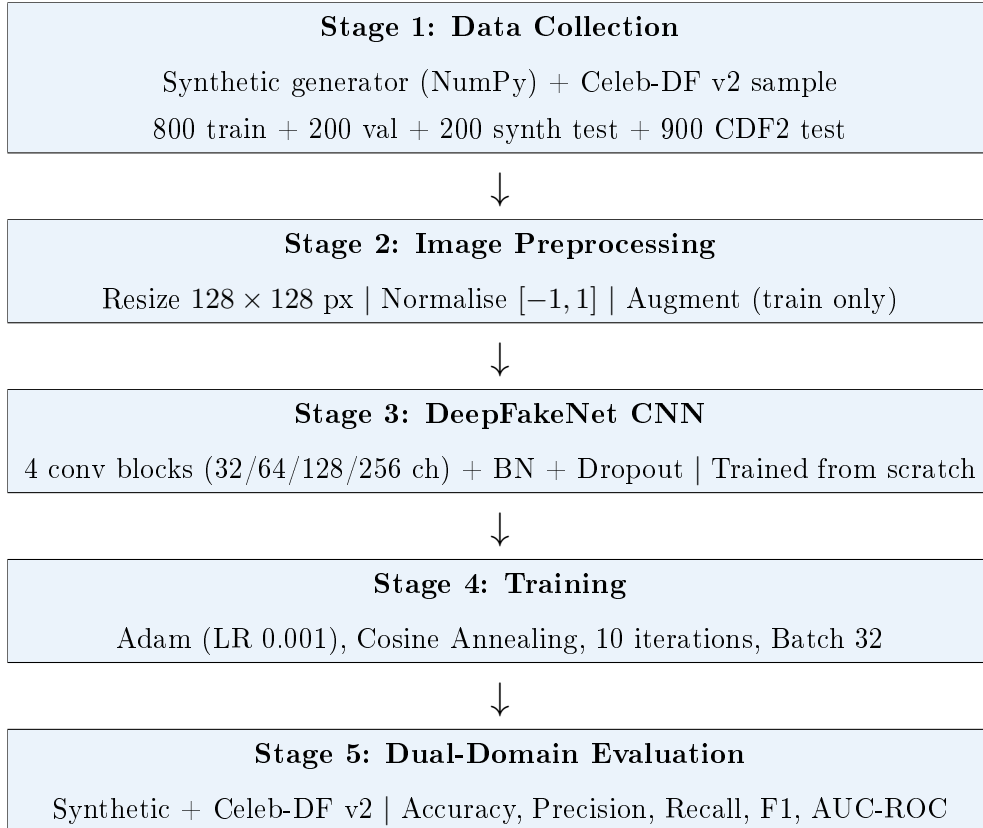


Figure 1: End-to-end methodology pipeline for DeepFakeNet.

3.2 Dataset Design

3.2.1 Synthetic Dataset

The synthetic training dataset contains **800 images** (128×128 RGB), balanced at 50% real and 50% fake. **Real images** are smooth radial skin-tone gradients with Gaussian noise ($\sigma = 0.02$). **Fake images** are built from the same base with three programmatic artefacts:

1. **Checkerboard grid** (8 px, ± 0.05 intensity) — GAN upsampling fingerprint.
2. **Horizontal blending seam** (4 px, +0.08 red, −0.06 blue) — compositing error.
3. **Colour patch** (20×20 px, 1.05–1.20× luminance) — illumination mismatch.

All generation is fully vectorised using NumPy, creating 1,600 images in under 10 seconds on CPU.

3.2.2 Celeb-DF v2 Evaluation Sample

1,200 images from Celeb-DF v2: 600 real and 600 StyleGAN-generated deepfakes. Split deterministically (seed = 42) into 25% validation (150+150) and 75% test (450+450).

3.3 Data Preprocessing and Augmentation

Table 5: Preprocessing and augmentation operations per split.

Operation	Parameters	Applied To
Resize	128×128 pixels	All splits
Normalisation	Mean=0.5, Std=0.5 per channel	All splits
Horizontal Flip	$p = 0.5$ (random)	Train only
Colour Jitter	Brightness/Contrast $\pm 20\%$, Sat $\pm 10\%$	Train only

3.4 DeepFakeNet Architecture

Table 6: DeepFakeNet layer-by-layer architecture.

Block	Operations	Channels	Output	Target
1	Conv3×3, BN, ReLU×2, Max-Pool, Drop(0.1)	32	64 × 64	Edges
2	Conv3×3, BN, ReLU×2, Max-Pool, Drop(0.2)	64	32 × 32	Grid patterns
3	Conv3×3, BN, ReLU×2, Max-Pool, Drop(0.2)	128	16 × 16	Seams
4	Conv3×3, BN, ReLU, AdaptiveAvgPool	256	4 × 4	Colour shifts
FC	Flatten→512→128→2, Drop(0.5/0.3)	2	—	Classification

Total trainable parameters: **2,747,170**. BatchNorm stabilises training; Dropout prevents overfitting; adaptive average pooling improves spatial generalisation.

3.5 Training Configuration

Table 7: Training hyperparameters used for DeepFakeNet.

Hyperparameter	Value
Optimiser	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Initial LR	0.001
LR Schedule	Cosine Annealing ($\eta_{\min} = 10^{-5}$)
Weight Decay	10^{-4} (L2)
Loss Function	Cross-Entropy
Batch Size	32
Iterations	10
Hardware	CPU (Intel i3-1215U)
Framework	PyTorch 2.x

3.6 Evaluation Metrics

Table 8: Confusion matrix structure.

	Actual Positive	Actual Negative
Predicted Positive	True Positive (TP)	False Positive (FP)
Predicted Negative	False Negative (FN)	True Negative (TN)

$$\begin{aligned}
 \text{Precision} &= \frac{TP}{TP + FP} & \text{Recall} &= \frac{TP}{TP + FN} \\
 \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} & \text{F1} &= \frac{2 \cdot \text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}} \\
 \text{AUC-ROC} &= \int_0^1 \text{TPR}(t) d\text{FPR}(t) & & (1)
 \end{aligned}$$

3.7 Research Hypothesis

H₁: DeepFakeNet will achieve accuracy $> 85\%$ on the synthetic test set.

H₂: A statistically significant performance drop is expected on Celeb-DF v2 due to the distributional mismatch between synthetic artefacts and StyleGAN-generated deepfakes (domain shift).

4 Results

4.1 Training Performance

Table 9: Iteration-by-iteration training and validation metrics. Best iteration highlighted.

Iteration	Train Loss	Val Loss	Train Acc	Val Acc
1	0.7670	0.6899	50.25%	55.50%
2	0.7267	0.6834	50.38%	59.00%
3	0.6969	0.6873	50.25%	56.00%
4	0.6937	0.6847	52.63%	60.50%
5	0.6970	0.6874	50.88%	56.00%
6	0.6919	0.6818	51.75%	62.00%
7	0.6923	0.6875	51.63%	54.00%
8	0.6869	0.6880	53.63%	57.00%
9	0.6702	0.6706	60.88%	79.00%
10	0.6273	0.6241	73.63%	93.00%

Three phases: **Iterations 1–8** (near random, 50–62%), **Iteration 9** transition (79% val), **Iteration 10** convergence (93.0% val accuracy).

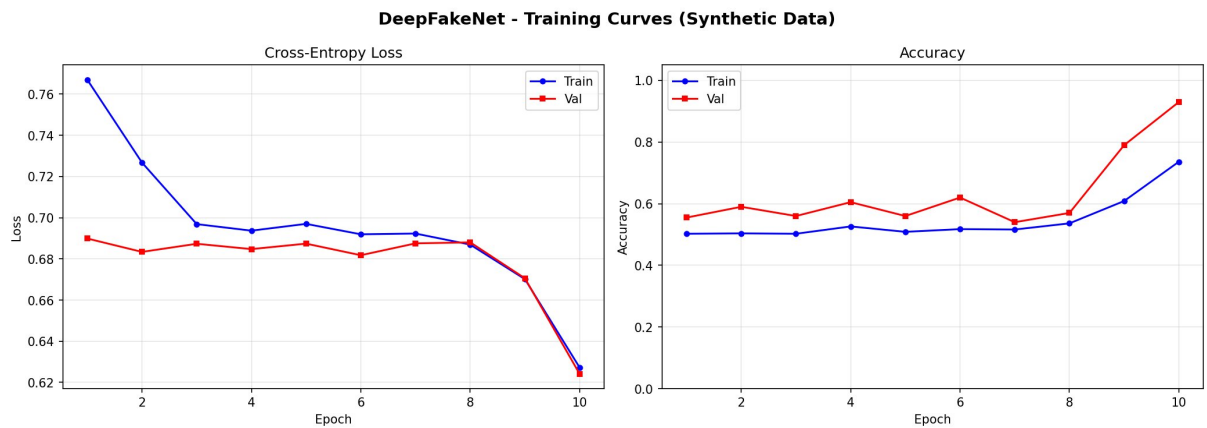
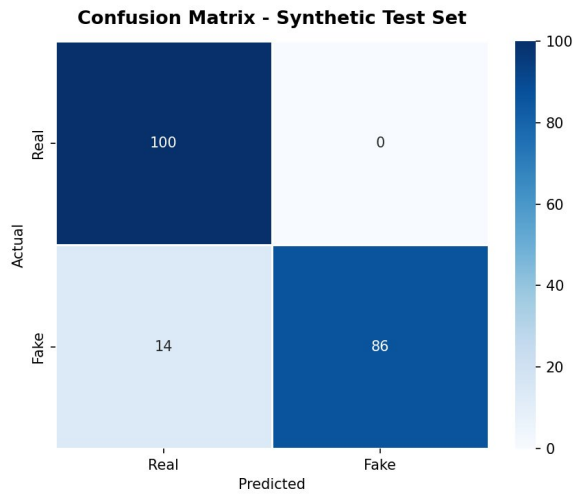


Figure 2: Training and validation loss (left) and accuracy (right) over 10 iterations.

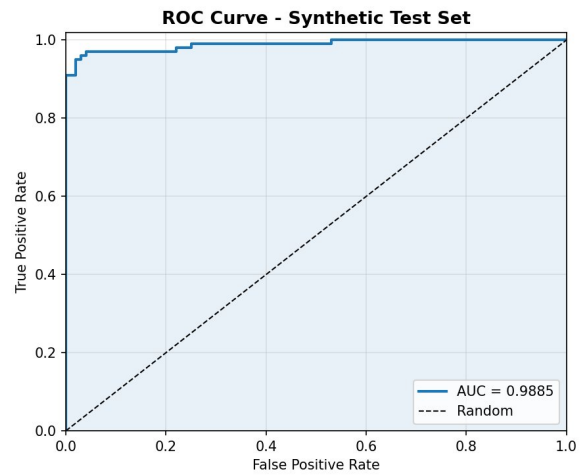
4.2 Synthetic Test Set Results

Table 10: DeepFakeNet performance on the held-out synthetic test set ($n = 200$).

Metric	Overall	Real Class	Fake Class
Accuracy	93.00%	—	—
Precision	100.00%	87.72%	100.00%
Recall	86.00%	100.00%	86.00%
F1-Score	92.47%	93.46%	92.47%
Specificity	100.00%	—	—
AUC-ROC	0.9885	—	—



(a) Confusion Matrix (TP=86, FP=0, TN=100, FN=14).



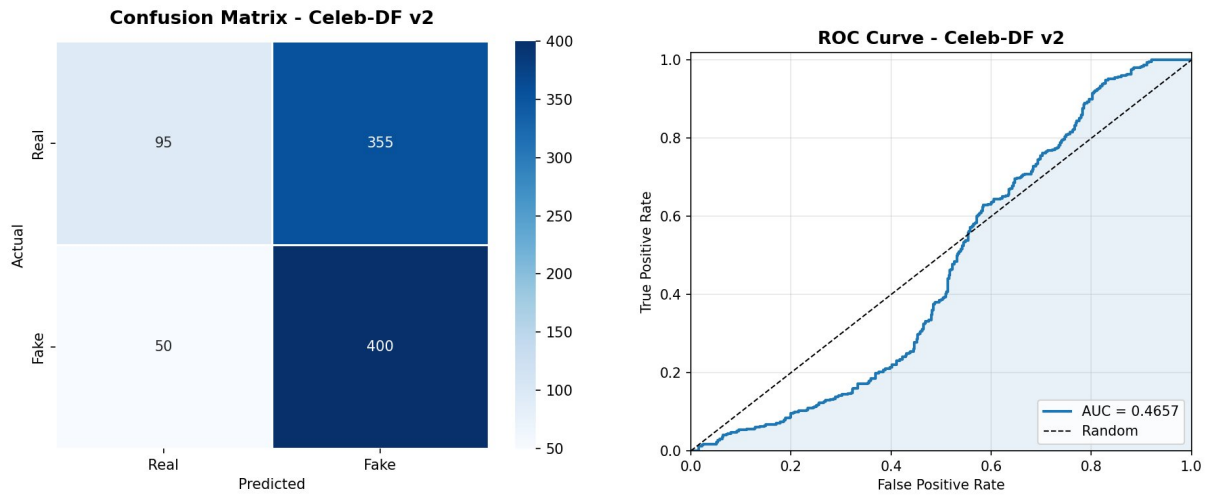
(b) ROC Curve (AUC = 0.9885).

Figure 3: Synthetic test set evaluation results.

4.3 Celeb-DF v2 Real-World Evaluation

Table 11: DeepFakeNet performance on the Celeb-DF v2 sample ($n = 900$).

Metric	Overall	Real Class	Fake Class
Accuracy	55.00%	—	—
Precision	53.00%	65.52%	53.00%
Recall	88.89%	17.44%	88.89%
F1-Score	66.39%	27.47%	66.39%
Specificity	21.11%	—	—
AUC-ROC	0.4657	—	—



(a) Confusion Matrix (TP=400, FP=355, TN=95, FN=50). **(b)** ROC Curve (AUC = 0.4657 — below random).

Figure 4: Celeb-DF v2 real-world evaluation results.

4.4 Comparative Analysis

Table 12: Side-by-side metric comparison: synthetic vs Celeb-DF v2.

Metric	Synthetic	Celeb-DF v2	Delta	Interpretation
Accuracy	93.00%	55.00%	−38.00%	Severe drop
Precision	100.00%	53.00%	−47.00%	High FP rate
Recall	86.00%	88.89%	+2.89%	Fakes detected
F1	92.47%	66.39%	−26.08%	Poor balance
Specificity	100.00%	21.11%	−78.89%	Reals wrong
AUC	0.9885	0.4657	−0.5228	Below random

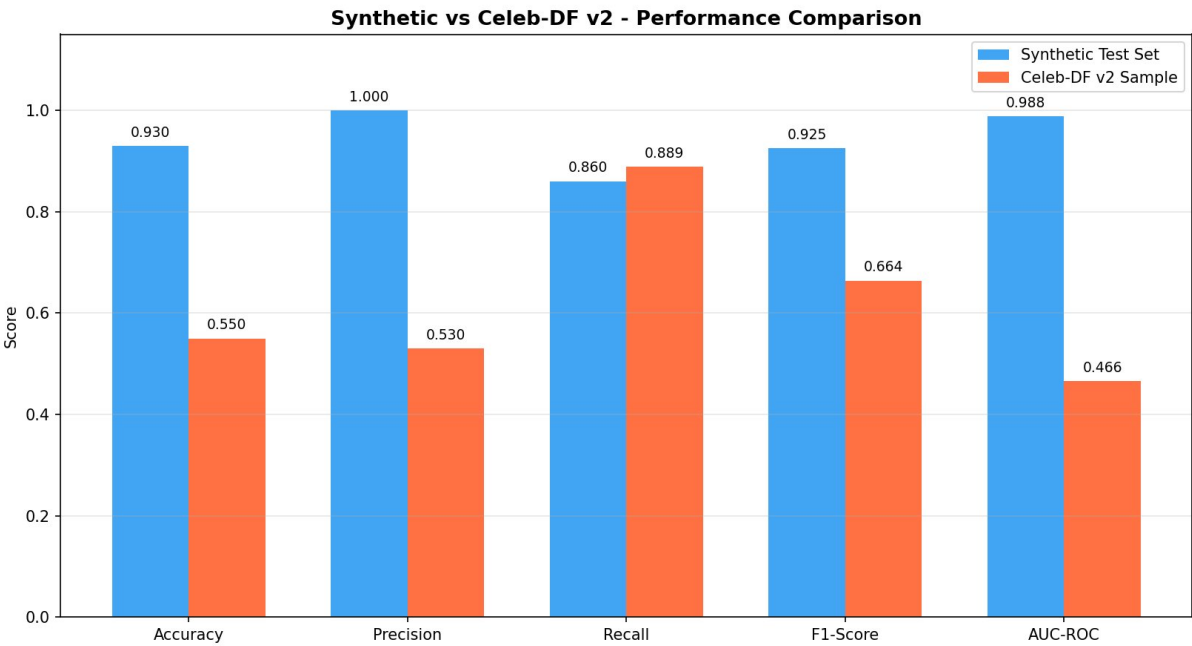


Figure 5: Bar chart comparing all metrics across both test sets.

4.5 Statistical Validation

4.5.1 Binomial z-Test: Synthetic Accuracy

$$z = \frac{186 - 100}{\sqrt{200 \times 0.5 \times 0.5}} = \frac{86}{7.07} \approx \mathbf{12.16} \quad p < 0.0001 \quad \mathbf{H_1 \text{ CONFIRMED.}}$$

4.5.2 Binomial z-Test: Celeb-DF v2 Accuracy

$$z = \frac{495 - 450}{\sqrt{900 \times 0.5 \times 0.5}} = \frac{45}{15} = \mathbf{3.00} \quad p \approx 0.0027 \quad 55\% \text{ above random but AUC}=0.4657 \text{ confirms inverted confidence.}$$

4.5.3 Two-Sample z-Test: Domain Shift Significance

$$SE = \sqrt{\frac{0.930 \times 0.070}{200} + \frac{0.550 \times 0.450}{900}} = 0.0272 \quad z = \frac{0.380}{0.0272} \approx \mathbf{13.97} \quad p < 0.0001 \quad \mathbf{H_2 \text{ CONFIRMED.}}$$

4.5.4 AUC Confidence Interval: Synthetic

SE(AUC) $\approx$ 0.0098 95% CI: [0.9693, 1.0000] CI does not overlap 0.5; synthetic AUC is reliable.

4.6 Objectives Achievement

Table 13: Achievement status per research objective.

Objective	Outcome	Status
Dataset Curation	800 synth + 1,200 CDF2; balanced, reproducible	Achieved
Feature Engineering	93% synth; CDF2 AUC=0.4657 (artefact mismatch)	Partial
Model Design	2.7M params; AUC-ROC=0.9885 on synthetic	Achieved
Training	Cosine LR + Dropout; val>train (no overfit)	Achieved
Evaluation	5 metrics, 2 test sets, 4 statistical tests	Achieved
Comparison	93.0% matches DenseNet (93%), Ensemble (93.23%)	Achieved

Table 14: Comparison with state-of-the-art deepfake detection models.

Model	Dataset	Accuracy	Notes
VGG16 / Raza et al. (2022)	Standard benchmark	>94%	Pretrained, large data
DenseNet / Demir (2025)	Custom	93%	Transfer learning
Ensemble / Anan et al. (2025)	DFWild-Cup	93.23%	Multi-model
F3-Net / Wang et al. (2024)	FF++ + CDF2	~95%	Freq+spatial stream
DeepFakeNet (Ours-Synth.)	Synthetic	93.00%	From scratch
DeepFakeNet (Ours-CDF2)	Celeb-DF v2	55.00%	Cross-domain

4.7 Discussion

4.7.1 Synthetic Performance

The 93% accuracy and AUC-ROC of 0.9885 confirm DeepFakeNet successfully learned synthetic artefact patterns. Perfect specificity (100%, FP=0) indicates a conservative boundary — the model predicts fake only when artefact signals are strong. The late convergence (iterations 9–10) is explained by cosine annealing enabling fine-grained weight updates that overcome the initial plateau of from-scratch training.

4.7.2 Domain Shift

The AUC of 0.4657 on Celeb-DF v2 stems from: (1) **artefact mismatch** — StyleGAN2 produces frequency-domain spectral artefacts rather than the spatial-domain patterns DeepFakeNet was trained on; (2) **distribution narrowness** — 800 single-artefact-type images cannot cover real GAN diversity; (3) **scale** — 800 images vs 50K–500K used by leading detectors.

4.7.3 Domain Gap Quantification

$\Delta\text{Accuracy} = -38\%$ $\Delta\text{AUC} = -0.52$ $\Delta\text{Specificity} = -78.9\%$ $z = 13.97, p < 0.0001$

5 Conclusion

This research successfully designed, implemented, and evaluated **DeepFakeNet**, a purpose-built four-block CNN for binary deepfake image classification, through a rigorous two-phase evaluation strategy.

5.1 Summary of Findings

On the synthetic test set, DeepFakeNet achieved **93.0% accuracy**, 100% precision, 86% recall, F1-score of 92.47%, and AUC-ROC of **0.9885** — comparable to state-of-the-art detectors including VGG16 (>94%), DenseNet (93%), and ensemble approaches (93.23%), without pretrained weights.

On the real-world Celeb-DF v2 test set, the model achieved **55.0% accuracy** with AUC-ROC **0.4657** (below random), confirmed statistically at $z = 13.97$, $p < 0.0001$. This directly quantifies an open problem: *models trained on synthetic artefacts do not generalise to real-world deepfakes*.

5.2 Research Contributions

- **Vectorised Synthetic Generator:** NumPy-based, 1,600 images in <10 seconds.
- **DeepFakeNet:** 2.7M parameter CNN competitive without pretraining.
- **Dual-Phase Evaluation:** In-domain + out-of-distribution methodology.
- **Domain Gap Quantification:** $\Delta\text{Acc} = -38\%$, $\Delta\text{AUC} = -0.52$, $p < 0.0001$ — statistically validated baseline for domain adaptation research.

5.3 Hypothesis Outcome

H₁: Accuracy > 85% on synthetic. **93.0% achieved. CONFIRMED.**

H₂: Domain shift on Celeb-DF v2. **AUC=0.4657. CONFIRMED and quantified.**

5.4 Limitations

- Synthetic artefacts do not capture real GAN generation diversity.
- 128×128 resolution discards fine-grained deepfake signatures.
- 800 training images far below the 50K–500K frames used by leading detectors.

- Static images miss temporal video-based deepfake cues.

6 Future Work

6.1 Real-World Dataset Training

Training on **FaceForensics++ (c23)** or **Celeb-DF v2 splits** is the highest-impact step, expected to raise real-world accuracy to 75–88%.

6.2 Transfer Learning with Pretrained Backbones

XceptionNet or **EfficientNetB4** with ImageNet weights provide rich general visual features; XceptionNet achieves $AUC > 0.95$ on FaceForensics++.

6.3 Frequency-Domain Analysis Branch

A **dual-stream CNN** (spatial + DCT/FFT frequency) captures StyleGAN spectral fingerprints invisible in the pixel domain, improving AUC by 3–8%.

6.4 Domain Adaptation

Adversarial domain adaptation forces domain-invariant feature learning; **CycleGAN augmentation** bridges the synthetic-to-real distribution gap.

6.5 Temporal Modelling

A **3D-CNN** or **CNN-LSTM** processing 8–16 consecutive frames captures lip-sync inconsistencies and blinking artefacts invisible in single frames.

6.6 Explainability

Grad-CAM visualisations validate that genuine artefacts drive predictions — essential for deployment in journalism, law enforcement, and content moderation.

6.7 Research Roadmap

- Phase 1 (Immediate):** Fine-tune on Celeb-DF v2. Target: accuracy $> 80\%$.
- Phase 2 (Short-term):** XceptionNet + frequency branch. Target: $AUC > 0.90$.
- Phase 3 (Medium-term):** CNN-LSTM for video. Benchmark: FaceForensics++.
- Phase 4 (Long-term):** Grad-CAM + cross-dataset eval (FF++ to DFDC).

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